

Posterior Shoulder Instability

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Posterior shoulder instability is a unique condition that not only represents a spectrum of instability, but can also be difficult to diagnose and technically challenging to treat. Although less common than anterior shoulder instability, accounting for only 2% to 10% of cases, this condition is becoming increasingly recognized in the athletic population.¹⁻⁴ McLaughlin et al. first recognized the wide clinical spectrum of posterior shoulder instability ranging from the locked posterior shoulder dislocation to recurrent posterior subluxation (RPS).² The causes are multifactorial and include acute traumatic injury, repetitive microtrauma, and atraumatic or ligamentous laxity.⁵ In the general population, seizures and electrocution are rare causes of posterior shoulder dislocation. More commonly in the traumatic setting, a posterior direct blow to an adducted, internally rotated and forward flexed upper extremity such as a fall on an outstretched arm is the mechanism of injury.⁶ In the athletic population, however, instability results from repetitive microtrauma, which can occur in a variety of loading conditions and multiple arm positions.^{2,7} Management of patients with posterior shoulder instability is challenging in recognition of the diagnosis and underlying pathology. Treatment includes both conservative and operative management. Surgeries for posterior instability include both open and arthroscopic procedures, with arthroscopic procedures recently gaining more popularity with advances in surgical techniques and decreased morbidity.

Background

RPS has been observed in overhead athletes, baseball hitters, golfers, tennis players, butterfly and freestyle swimmers, weightlifters, rugby players, and football linemen.⁸⁻¹⁰ Other activities where RPS may be present include archery, riflery, and wheelchair use.¹¹ Distinct from the acute traumatic dislocation, the etiology of RPS is due to repetitive microtrauma to the posterior capsule leading to capsular attenuation and labral tears. In the overhead athlete, adaptive and structural changes in combination with the high forces across the glenohumeral joint result in posterior capsular contracture and posterosuperior labral tear, which is a distinct mechanism from other entities.

Diagnosis and workup of athletes with RPS is often challenging when determining the underlying pathology. These patients present with complaints of vague and diffuse shoulder pain and fatigue without a specific injury. A thorough history and physical and examination in combination with specific imaging studies are required to determine the pathogenesis and treatment for

patients with RPS. Provocative clinical examination tests in addition to advanced imaging such as the MR arthrogram have aided in identifying RPS in the athletic population. A thorough history should include the mechanism of injury, which can consist of a true posterior shoulder dislocation, repetitive microtrauma, acute, or chronic subluxation. Specific direction of instability, meaning posterosuperior, directly posterior, posteroinferior as well as pattern of instability, including unidirectional or multidirectional, should be identified. All of these factors should be documented and considered, as they will ultimately affect treatment and outcome.¹² Structural abnormalities of the shoulder, to include any combination of the labrum, capsule, supporting ligaments, or rotator cuff, also must be taken in account to maximize optimal treatment outcome.

Posterior glenohumeral stabilization has evolved from various open procedures to an all arthroscopic approach.¹³ This has allowed for enhanced identification of repair of intra-articular pathology.¹⁴ As outcomes have been improving, clinical and biomechanical studies continue to refine our approach to RPS.

Pathogenesis

The posterior labrum, capsule, and posterior band of the inferior glenohumeral ligament (IGHL) are the primary stabilizers to posterior translation of the humeral head between 45 and 90 degrees of abduction.¹⁵ The posterior capsule is identified by the area between the intra-articular portion of the biceps tendon and the posterior band of the IGHL. This area is the thinnest segment of the shoulder capsule and also does not contain any supporting ligamentous structures, making it susceptible to attenuation from repetitive stresses.¹⁶

The pathogenesis of RPS shares a direct relationship to the specific repetitive stresses placed onto the glenohumeral joint. This differs from the pathogenesis in a traumatic setting where the patient may sustain an acute capsulolabral detachment or reverse Bankart tear (Figs. 41.1 and 41.2).¹⁷ In RPS, specific activities such as the tennis backhand, golf or baseball backswing, follow-through phase of throwing, and pull-through phase in swimming all can result in RPS. Activities such as push-ups, bench press, and blocking in football also place a direct posterior stress on the posterior capsulolabral complex of the shoulder. Due to the various positions athletes use in contact and non-contact sport, many other mechanisms likely exist that are continuously being identified.

Abstract

Posterior shoulder instability is a unique condition that not only represents a spectrum of instability, but can also be difficult to diagnose and technically challenging to treat. Although less common than anterior shoulder instability, accounting for only 2% to 10% of cases, this condition is becoming increasingly recognized in the athletic population.¹⁻⁴ McLaughlin et al. first recognized the wide clinical spectrum of posterior shoulder instability ranging from the locked posterior shoulder dislocation to recurrent posterior subluxation (RPS).² The causes are multifactorial and include acute traumatic injury, repetitive microtrauma, and atraumatic or ligamentous laxity.⁵ In the general population, seizures and electrocution are rare causes of posterior shoulder dislocation. More commonly in the traumatic setting, a posterior direct blow to an adducted, internally rotated and forward flexed upper extremity such as a fall on an outstretched arm is the mechanism of injury.⁶ In the athletic population, however, instability results from repetitive microtrauma, which can occur in a variety of loading conditions and multiple arm positions.^{2,7} Management of patients with posterior shoulder instability is challenging in recognition of the diagnosis and underlying pathology. Treatment includes both conservative and operative management. Surgeries for posterior instability include both open and arthroscopic procedures, with arthroscopic procedures recently gaining more popularity with advances in surgical techniques and decreased morbidity.

Keywords

posterior shoulder instability
posterior shoulder instability reconstruction
posterior shoulder instability repair
arthroscopic posterior capsulolabral reconstruction
recurrent posterior shoulder instability

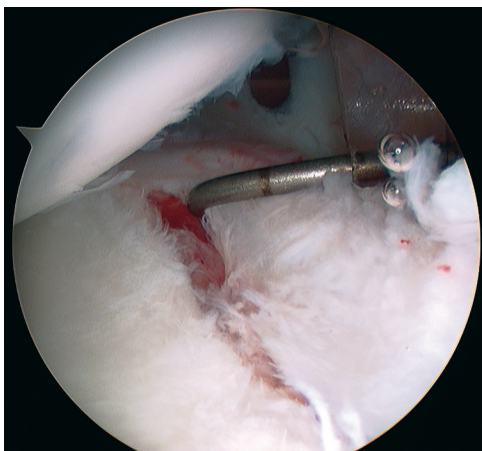


Fig. 41.1 A posterior labral tear from the glenoid (right shoulder, viewed from the anterior portal).

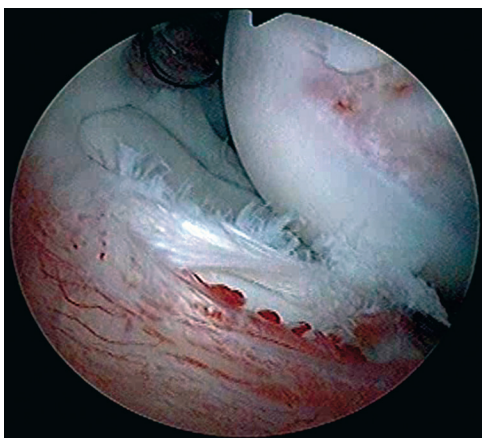


Fig. 41.2 Posterior labral splitting and tear (right shoulder, viewed from the posterior portal). Placing the patient in the lateral decubitus position facilitates posterior labral visualization and repair.

One proposed mechanism of RPS in overhead athletes, throwers, and swimmers is due to the progressive laxity of the posterior capsule and fatigue of static and dynamic stabilizers.¹⁸ RPS can lead to deformation of the posterior capsule, resulting in a patulous posteroinferior capsular pouch that can be seen on both magnetic resonance arthrogram (MRA) and arthroscopically.

Cadaveric studies have examined position and applied force in relation to posterior instability. Pagnani et al. placed the arm into the position of flexion and internal rotation and found that transection of the infraspinatus, teres minor, and entire posterior capsule was sufficient to produce a dislocation.¹⁶ When the anterosuperior capsule and superior glenohumeral ligament (SHGL) were transected, a posterior dislocation would occur. These studies validated the “circle concept,” where a dislocation in one direction would require capsular damage on both the same and opposite side of the joint. A different study by Cole et al. found that with the arm in an adducted position, the rotator interval and SGHL provided static glenohumeral stability limiting inferior and posterior joint translations.¹⁹ Wellmann et al. found in their cadaveric study that a lesion of the rotator interval contributes to increased glenohumeral translation.²⁰ However,

Provencher and colleagues demonstrated that rotator interval closure, while causing loss of external rotation, did not have an effect on reducing posterior instability.²¹ Other anatomic studies have questioned the concept of injury to anterior structures, including the rotator interval in a posteriorly dislocated shoulder altogether.^{22,23} This work, alongside others, suggests that in an overhead athlete, interval closure may not be clinically indicated and may in fact have an undesired effect on throwing athletes who function with their arm in the abduction and external rotation (ABER) position.

The muscles around the shoulder also significantly contribute to the dynamic cavity compression effect of the humeral head within the glenoid.²⁴ Of the rotator cuff muscles, the subscapularis provides the most resistance to posterior translation, acting as a dynamic supporter of the action of the posterior band of the IGHL.^{25,26} In one cadaveric study, decreased subscapularis muscle strength in the late cocking phase of throwing motion resulted in increased maximum glenohumeral external rotation and increased glenohumeral contact pressure.²⁷ Therefore overhead throwers with subscapularis fatigue would be more susceptible to these forces, resulting in a type II superior labral anterior posterior (SLAP) tear. In some instances these tears can propagate posteroinferiorly around the glenoid resulting in type VIII SLAP tear and symptomatic RPS.^{27,28} From an anatomic standpoint, the humeral head possesses an oblong shape. In the ABER position the anterior band of the IGHL becomes taut as it is draped over the anteroinferior aspect of the eccentrically positioned humeral head, providing a check-rein against excessive external rotation.^{29,30}

Pitchers that have increased external rotation at the expense of internal rotation will have more rapid internal humeral rotation in their follow through and pitch velocity. Chronic use of the shoulder in this manner will result in a posterior capsular contracture and glenohumeral rotation deficit. Burkhart suggested that the oblong (cam) effect of the humerus is counteracted by the development of posterior capsular tightness, thus shifting the rotational fulcrum posterosuperiorly.³⁰ This shift allows the shoulder to clear the anteroinferior labroligamentous restraints while leading them to be susceptible to posterior and posterosuperior microtrauma events.^{4,18,31} This occurs during the follow-through phase with the shoulder in a flexed, adducted, and internally rotated position.

The posteroinferior capsular thickening associated with glenohumeral internal rotation deficit (GIRD) has been visualized on magnetic resonance imaging (MRIs) of throwing athletes.³² Dynamic MRA with the arm in the ABER position has shown a phenomenon known as posterosuperior labral “peel-back.” On MRA with the arm in the ABER position, the detached posterosuperior labral tissue separates and moves medially toward the glenoid rim. This can also be visualized arthroscopically (Fig. 41.3).²⁸

It has been observed that some adaptive changes can occur in the shoulder without negative consequence, whereas other athletes may go on to become symptomatic. Studies have demonstrated that after approximately 25 degrees of GIRD, the posterosuperior shift in the glenohumeral center of rotation progresses from adaptive to pathologic shearing forces on the

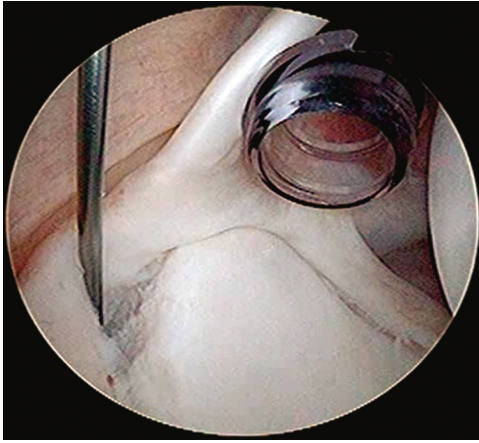


Fig. 41.3 Complete detachment of the posterosuperior labrum from the glenoid in an overhead thrower, as localized by the spinal needle (right shoulder, viewed from the posterior portal).

posterosuperior labrum and rotator cuff.³³ Excessive external rotation in the ABER position exposes the posterosuperior labrum, longhead of the biceps tendon, and undersurface rotator cuff tendons to excessive rotary forces resulting in peel-back phenomenon.³⁴ Posterior shift of the humeral head in the ABER position also results in a relative redundancy of the anteroinferior capsuloligamentous complex due to a decreased cam effect. Although not the primary pathology of a tight posteroinferior capsule, this secondary anterior pseudolaxity may have clinical implications in the pathophysiology of the disabled throwing shoulder.

The presentation of a posterosuperior labral tear in combination with posteroinferior capsular contracture in throwing athletes differs from the spectrum of pathology in lifters, offensive lineman, and athletes who receive repetitive high loads with the arm in the forward flexed and adducted position.⁹ The pathogenesis of these conditions is important because initial treatment may differ based on the underlying etiology. Initial treatment in overhead athletes with GIRD and RPS is conservative with posteroinferior capsular contracture stretching, “sleeper stretches” is a more appropriate initial treatment.^{30,35} Subscapularis treatment may also play a role. If the thrower remains symptomatic, arthroscopic posterior labral repair with optional capsular release can then be performed.

“Batter’s shoulder” is another cause of RPS in baseball hitters where this syndrome affects the lead shoulder during the baseball swing.^{36,37} As the dynamic pulling forces approach 500 Newtons, a posterior labral tear results. It has been inferred that golfers may experience a similar mechanism of injury along with any athlete with a similar follow through type motion during swinging.

Identification and treatment of scapular dyskinesia is also of importance when identifying the pathogenesis of RPS. Scapular winging can act as a compensatory mechanism to prevent posterior subluxation of the humeral head. In other patients, winging is thought to be the primary etiology of subluxation.⁴ A study in golfers with posterior shoulder instability found that fatigue developed in the serratus anterior muscle.³⁸ This possibly contributing to scapulothoracic asynchrony and the combination

of posterior shoulder instability and subacromial impingement. An initial treatment in these patients should include a periscapular muscle strengthening program, which should be incorporated into the physical therapy protocol.

HISTORY

Patients with RPS often have vague and nonspecific complaints and symptoms that make clinical diagnosis difficult to elucidate. Symptoms can include any combination of activity related pain, crepitus, perceived weakness, and episodes of intermittent subluxation. If capsular contracture has developed, a complaint of loss of motion may also be present. One study by Pollock et al. noted that two-thirds of athletes who ultimately required surgery presented with difficulty using the shoulder outside of sports with use of the arm above the horizontal.³⁹ Another study found that 90% of patients with RPS note clicking or crepitation with motion.

PHYSICAL EXAMINATION

A thorough physical examination of the shoulder includes initial inspection focusing on shoulder asymmetry, muscular atrophy, or scapular dysrhythmia. A skin dimple over the posteromedial deltoid of both shoulders has been found to be 62% sensitive and 92% specific in correlating with posterior instability.⁴⁰

On palpation, tenderness is assessed at the posterior glenohumeral joint line, greater tuberosity, and biceps tendon. In patients with RPS, posterior joint line tenderness is likely a result of posterior synovitis or posterior rotator cuff tendinosis secondary to multiple episodes of instability.³⁹

Shoulder range of motion (ROM) should also be assessed in both shoulders. Record measurements of forward elevation, abduction in the scapular plane, external rotation with the arm at the side, and internal rotation with the hand behind the back to the highest vertebral level. Shoulder measurements while the patient is in the supine position should also be recorded. While supine, the arm is abducted 90 degrees, the scapula is stabilized, and glenohumeral external and internal rotation are measured. Total arc of rotation is calculated by adding total external plus total internal rotation, and GIRD is assessed by assessing the side to side difference in internal rotation. Crepitus can oftentimes be reproduced during ROM examination as well.

Strength testing is performed bilaterally, with a focus on the rotator cuff. Grading is performed on a five-point scale ([Table 41.1](#)). The majority of athletes with RPS will demonstrate grade 4 or 5 strength. Supraspinatus strength is tested with a downward force while the arm is abducted 90 degrees in the scapular plane, otherwise known as the “empty can” test. Infraspinatus and teres minor muscles are assessed with resisted external rotation, with the arm adducted and the elbow flexed 90 degrees. If the posterior cuff has been damaged, subtle differences can be detected. Subscapularis strength testing is performed with the lumbar lift-off, belly press, and bear hug tests.^{41–43}

Glenohumeral stability is assessed in both shoulders with the patient supine. The load and shift maneuver is performed with the arm held in 90 degrees of abduction and neutral rotation.

TABLE 41.1 Muscle Grading Scale

0	No muscle contraction is detectable
1	A contraction can be seen or palpated but strength is insufficient to move the joint at all
2	The muscle can move the joint if the limb is oriented so that the force of gravity is eliminated
3	The muscle can move the joint against the force of gravity, but not against any additional applied force
4	The muscle can move the joint against the force of gravity and additional applied force but is not believed to be normal
5	The muscle strength is considered normal

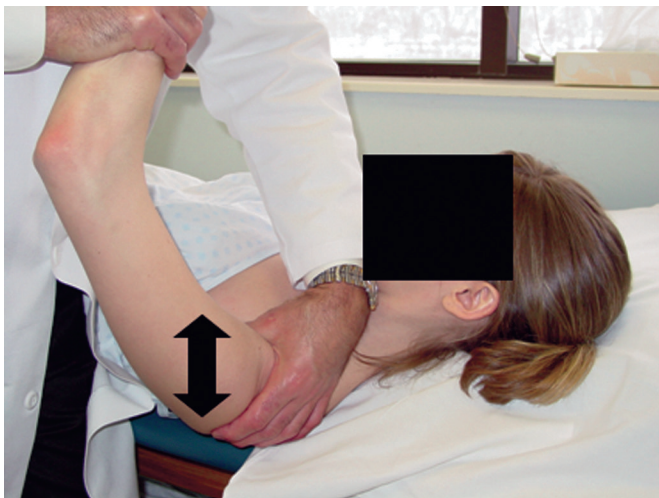


Fig. 41.4 The load-and-shift test is performed to assess anterior and posterior glenohumeral stability.

TABLE 41.2 Grading of Anterior and Posterior Laxity

0	A humeral head that does not translate to the glenoid rim
1+	A humeral head that translates to but does not translate over the glenoid rim
2+	A humeral head that translates over the glenoid rim but spontaneously reduces
3+	A humeral head that translates over the glenoid rim and does not spontaneously reduce

Axial load is applied to the glenohumeral joint with an anterior-directed force applied then a posterior directed force (Fig. 41.4). Translation of the humeral head over the anterior glenoid rim is quantified from 0 to 3+ (Table 41.2).

The sulcus test is performed to asses for excessive IGHL and SGHL laxity. With the patient in a seated position, longitudinal traction is applied with the arm adducted and in neutral rotation. The test is repeated in 30 degrees of external rotation. Laxity is quantified based on the acromial humeral distance (Table 41.3). A sulcus sign that remains 2+ or greater in 30 degrees of external rotation is considered pathognomonic for multidirectional instability (MDI). Patients who present with RPS and concomitant MDI may need to have the rotator interval

TABLE 41.3 Sulcus Test

1+	Acromiohumeral distance <1 cm
2+	Acromiohumeral distance between 1 and 2 cm
3+	Acromiohumeral distance >2 cm

A 3+ sulcus sign that remains 2+ or greater in 30 degrees of external rotation is considered pathognomonic for multidirectional instability.

TABLE 41.4 Beighton Score for Joint Hypermobility (9 Points)

Joint	Negative	Unilateral	Bilateral
Ability to hyperextend the elbows beyond 10 degrees	0	1	2
Ability to passively touch the thumb to the adjacent forearm with the wrist in flexion	0	1	2
Ability to passively hyperextend the small finger metacarpophalangeal joint more than 90 degrees	0	1	2
Ability to hyperextend the knees beyond 10 degrees	0	1	2
Ability to touch the palms to the floor with feet together	0	1	1

Patients who score 5 points or more on the Beighton scale are considered ligamentously lax.⁴⁴

addressed at the time of surgery, in addition to the posterior capsulolabral complex for successful stabilization.

Generalized ligamentous laxity is also important to examine, as this may contribute to MDI. The Beighton scale assesses the spectrum of ligamentous laxity based on a 9-point scale (Table 41.4).⁴⁴ Patients with a score of 5 or more are considered ligamentously lax.

Special tests have also been applied to further identify capsulolabral pathology in the shoulder. The jerk test is used to assess posterior instability while in the seated position. The medial border of the scapula is stabilized with one hand while the other hand applies a posterior directed force to the 90 degree forward flexed, adducted, and internally rotated arm. The test is positive if posterior subluxation, dislocation, or pain and apprehension are present.^{7,45}

The Kim test assesses posteroinferior shoulder instability.⁴⁶ With the patient seated, the arm is placed into 90 degrees of abduction in the scapular plane and an axial load is applied. The arm is then forward elevated an additional 45 degrees, and a posteroinferior vector is placed across the glenohumeral joint. A sudden onset of posterior subluxation with pain is positive, with 97% sensitivity in eliciting a posteroinferior labral lesion.⁴⁶

The circumduction test is performed to assess patients with higher grades of chronic posterior instability. With the elbow in full extension, the arm is placed into 90 degrees of forward elevation and slight adduction. A posterior load is then applied, which subluxes or possibly dislocates the humeral head posteriorly. The arm is then circumducted with a combination of abduction and extension until the head reduces in the glenoid. A palpable

clunk as the head reduces into the glenoid is significant for a positive test. In patients with chronic posterior instability, this test can often be performed without pain or guarding.

In some cases of RPS, and primarily involving overhead throwers, a posterior labral tear is seen in combination with a superior labral tear resulting in a type VIII SLAP tear.²⁸ The active compression (O'Brien) test for superior labral pathology should also be performed. With the patient seated, the arm is forward elevated 90 degrees, adducted 10 degrees, and internally rotated with the elbow extended; a downward force is placed on the arm. A positive test is when deep pain within the shoulder joint is elicited when the arm is in this position compared with when the arm is externally rotated and the same force is applied. It is important to note that deep pain in the shoulder joint is suggestive of superior labral pathology, and is different than pain at the acromioclavicular joint, which can result in a false positive result.

Impingement signs may also be present on physical examination. This impingement is typically due to underlying scapular dyskinesia. Stress-related changes in the posterior rotator cuff can manifest as a secondary impingement syndrome.

IMAGING

Initial radiographs of the shoulder should be obtained in the workup of RPS. Radiographs should consist of an anteroposterior view in the plane of the scapula, an axillary view, and a scapular Y lateral view. In most patients with RPS, these radiographic images are normal. In some cases, a posterior glenoid lesion or impaction of the anterior humeral head (reverse Hill-Sachs lesion) can be seen.^{44,45} The axillary view is particularly useful in assessing glenoid version. A West Point radiographic view can be useful detecting fractures of the glenoid rim or subtle ectopic bone formation around the glenoid.^{47,48} CT Scan can also be useful in providing osseous detail in posttraumatic or chronic cases.

MRA is the most sensitive diagnostic test for identifying lesions of the posterior labrum and capsule.⁴⁹ MRA findings indicative of posterior shoulder instability include posterior translation of the humeral head relative to the glenoid, posterior labrocapsular avulsion, posterior labral tear or splitting, discrete posterior capsular tears or rents, reverse humeral avulsion of the glenohumeral ligaments, posterior labrum periosteal sleeve avulsion, and subscapularis tendon avulsion (Fig. 41.5).^{50,51} The Kim classification is used to describe posterior labral tear morphology (Figs. 41.6–41.8; Table 41.5).⁵² Arthroscopically the Kim lesion appears as a crack at the junction of the posteroinferior glenoid articular cartilage and labrum through which a complete detachment of the deeper labrum from the glenoid rim can be identified. MRA can also be used to identify chondrolabral and glenoid version.⁵³ Kim examined 33 shoulders with atraumatic posterior instability, and when compared with age matched controls, the affected shoulders were found to have a glenoid that was more shallow and also had more osseous and chondrolabral retroversion present in the middle and inferior glenoid (Fig. 41.9).⁵³ Studies performed by Bradley and colleagues have also found in athletes with RPS higher chondrolabral retroversion and glenoid bony retroversion when compared with controls.^{8,54} Another study by Tung et al. found 24 patients with RPS had MRA findings of

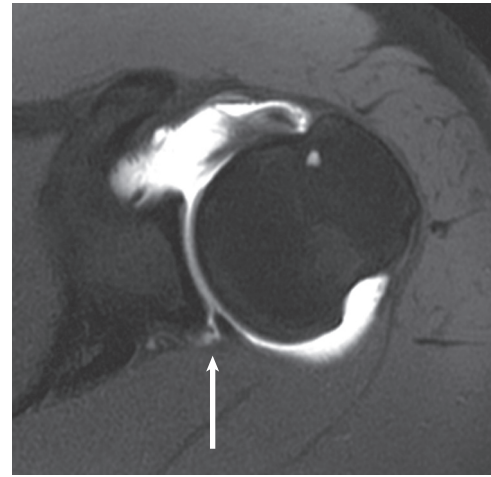


Fig. 41.5 An axial magnetic resonance imaging arthrogram image (left shoulder) demonstrating a posterior capsulolabral avulsion from the glenoid with associated posterior translation of the humeral head relative to the glenoid (arrow).

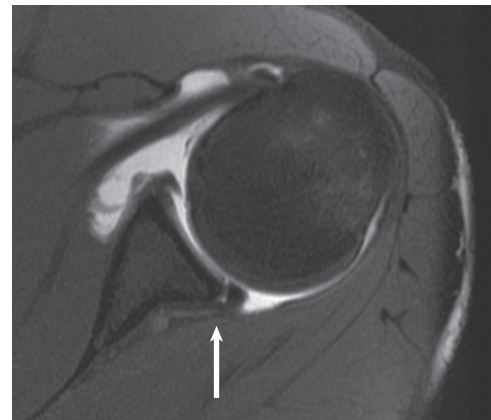


Fig. 41.6 An axial magnetic resonance imaging arthrogram image (left shoulder) demonstrating a type I Kim lesion: a posterior chondrolabral fissure without displacement (arrow).

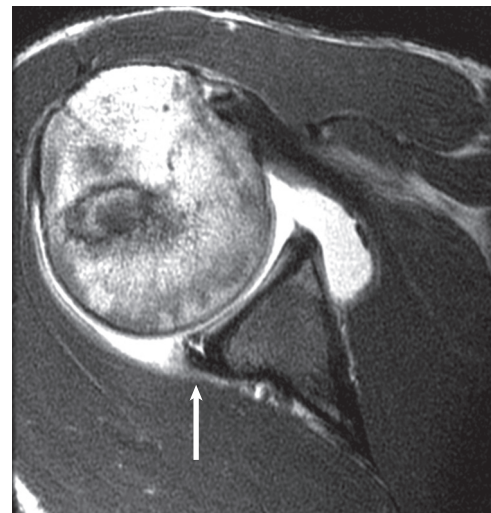


Fig. 41.7 An axial magnetic resonance imaging arthrogram image (right shoulder) demonstrating a type II Kim lesion: a concealed complete detachment of the posterior labrum from the glenoid (arrow).

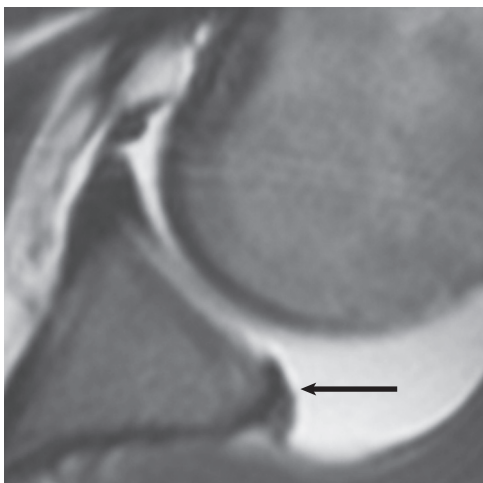


Fig. 41.8 An axial magnetic resonance imaging arthrogram image (left shoulder) demonstrating a type III Kim lesion: posterior chondrolabral erosion with loss of contour (arrow).

TABLE 41.5 Kim Classification of Posterior Labral Tear Morphology

Type I	Incomplete detachment (Fig. 41.6)
Type II (Kim Lesion)	Concealed complete detachment (Fig. 41.7)
Type III	Chondrolabral erosion (Fig. 41.8)
Type IV	Flap tear of the posteroinferior labrum

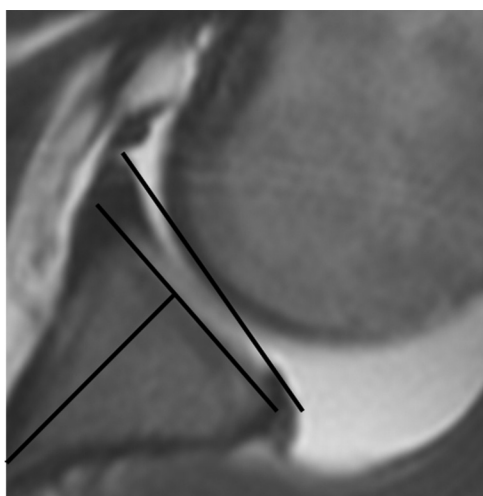


Fig. 41.9 An axial magnetic resonance imaging arthrogram image (left shoulder) demonstrating the technique used to measure the degree of chondrolabral retroversion, which can be increased in patients with recurrent posterior subluxation (RPS) of the shoulder.

more posterior humeral head translation, posterior labral tears, and posterior labrocapsular avulsions compared with healthier controls.⁴⁹ Dynamic MRA can also be performed to evaluate for the peel back phenomenon, which is consistent with a posterosuperior labral tear.⁵⁵ The use of CT is limited to cases in which a significant amount of bony glenoid retroversion is suspected.

DECISION-MAKING PRINCIPLES

Thorough knowledge of an athlete's sport, position, and training regimen is critical in determining the pathogenesis and specific pathology associated with their shoulder instability. Rehabilitation with an emphasis on strengthening the rotator cuff, posterior deltoid, periscapular muscles is commonly the first line of treatment.^{7,31,39,56–59} It is recommended that this physical therapy protocol be maintained for a minimum of 6 months to decrease an athlete's functional disability. A total of 70% of athletes with this protocol can return to sport. Although the RPS is not eliminated, the functional disability is improved for the athlete to participate in play. Therapy has been traditionally more effective in patients with atraumatic RPS, as opposed to those with generalized ligamentous laxity or a discrete traumatic event.^{31,57} For patients that fail therapy and are not able to return to preinjury levels of competition, surgery should be considered. Treatment options may ultimately include labral débridement, labral repair, posterior capsular release, posterior capsular plication, or a combination of these procedures.

TREATMENT OPTIONS

Many operative procedures have been described for the treatment of posterior instability, including both open and arthroscopic procedures. There has been a general trend in the transition from open procedures to arthroscopic repair, and from nonanatomic to an anatomic reconstruction. Open procedures included a reverse Putti-Platt procedure, biceps tendon transfer, subscapularis transfer, infraspinatus advancement, posterior opening glenoid wedge osteotomy, proximal humeral rotational osteotomy, bone block augmentation of the posterior glenoid or acromion, posterior staple capsulorrhaphy, allograft reconstruction, capsulolabral reconstruction, and open capsular shift.^{2,7,17,18,31,59–62}

Posterior bone block repair has been found to have good outcomes, with one study by Scapinelli reporting 100% success at 9.5 year follow-up.⁶³ However, a separate biomechanical study has shown that posterior bone block capsulorrhaphy overcorrects posterior translation and does not restore inferior glenohumeral stability.⁶⁴ Notable soft tissue tightening procedures include the reverse Putti-Platt and Boyd and Sisk procedures (rerouting the longhead of the biceps tendon to the posterior glenoid).^{3,65} In 1980 Neer and Foster reported good results in their laterally based posterior capsular shift to tighten a patulous posteroinferior capsule.⁶⁶ Other studies have attempted a medially based posterior capsular shift for posterior capsular tightening.^{18,31,59} Misamore et al. reported on 14 patients with traumatic unidirectional posterior instability who were treated with an open posterior capsulorrhaphy. They found excellent results, with 12 of 14 returning to preinjury level of play.⁵⁹ Other studies by Rhee presented a review of 33 shoulders for which open, deltoid saving posterior capsulolabral reconstruction was performed and reported that at 25 month follow-up, four patients had recurrent instability.⁶⁷ Another study by Wolf and colleagues presented similar findings in a retrospective review of 44 shoulders after open posterior glenohumeral stabilization.¹⁷ The recurrence rate was 19%, poor results in patients older than 37 years, and in those with chondral damage.¹⁷

Arthroscopic techniques for the treatment of shoulder instability have been implemented and become increasingly popular. Historically, arthroscopic thermal capsulorrhaphy was performed, but results have been fair at best. One study by Bisson presented 14 shoulders without labral detachment who underwent thermal capsulorrhaphy with 2-year follow-up.⁶⁸ They found that treatment failed in 3 of their 14 patients (21%). In another study, D'Alessandro found that 37% of patients who underwent thermal capsulorrhaphy for anterior, anteroinferior, or multidirectional shoulder instability had unsatisfactory results with American Shoulder and Elbow Surgeons (ASES) scores at 2- to 5-year follow-up.⁶⁹ Other studies by Miniaci reported 47% recurrent instability in patients being treated for MDI at 9 months post op.⁷⁰ Due to the variable response patients have demonstrated, thermal capsulorrhaphy for treatment of RPS is not recommended.

Arthroscopic capsulolabral repair has evolved over time, providing a minimally invasive approach with less morbidity and faster recovery.¹³ With advances in arthroscopic techniques, the all-arthroscopic approach to posterior stabilization has the ability to treat a wide array of pathology. With open techniques, anterior pathology of the shoulder (labrum tearing, subscapularis disruption), in addition to superior pathology (supraspinatus, biceps tendon, and superior labrum tearing), is near impossible to address through the same exposure. The arthroscopic approach allows for clear visualization of the entire labrum, rotator cuff, and other pain-generating pathology around the glenohumeral joint, which also allows for adequate treatment. In addition, arthroscopic management of posterior instability allows for complete labrum repair, in addition to capsular plication in an “around the world” fashion. Recent studies evaluating postoperative outcomes of the arthroscopic approach have also surpassed those of open techniques, making the arthroscopic modality the current standard of care for the majority of patients.

Indications

Patients are candidates for arthroscopic posterior stabilization if they present with prior episodes of posterior dislocation or RPS of the glenohumeral joint that has not responded favorably to conservative management. After an initial posterior dislocation occurs, a closed reduction is performed and instability is assessed. Once a pain-free shoulder with full ROM is restored, patients are gradually returned to regular activity. If resultant instability or pain persists and physiotherapy is not ameliorative, then the patient is counseled regarding surgical intervention.

Patients with RPS may not report an initial traumatic event; however, they may have an insidious course with deep shoulder pain, reduced athletic participation (inability to throw long ball, loss of velocity and control), and pain with cross body and adducted shoulder movements. An initial course of physical therapy can be initiated in addition to a throwing regimen for overhead-throwing athletes. If pain subsides, gradual return to play (RTP) is begun and the patient can be assessed for recurrence of symptoms.

If pain or subluxation returns, a MRA can be obtained to confirm the diagnosis. Once confirmed, the specific anatomical

considerations can be evaluated and the patient can be counseled regarding the need for surgical intervention. Primary indications for surgery include a history consistent with prior dislocation or RPS; physical examination that is positive for any or all of the provocative tests, including the Kim test, Jerk test, posterior load and shift, and pain with cross body adduction with reproduction of symptoms; diagnostic imaging (MRA) demonstrating a patulous posterior capsule, posterior labrum tear, capsular tear or concealed crack (“Kim” lesion); reverse humeral avulsion of the glenohumeral ligaments, reverse Hill-Sachs impaction fracture, reverse osseous Bankart lesion, relative increase in glenoid retroversion with hypoplastic labrum, and chondrolabral retroversion greater than 10 degrees; and patients who failed conservative treatment with a stretching and strengthening rotator cuff rehabilitation program and an inability to return to sport or activity at a preinjury level.

Contraindications

As arthroscopic techniques have evolved, so have the relative contraindications to an arthroscopic approach. Contraindications to arthroscopic surgery may include large glenoid (>25%) or reverse Hill-Sachs (>30%) lesions that may be better managed with an open bone grafting technique, capsular tearing with loss of capsular tissue (from prior surgery or thermal modification), excessive glenoid retroversion (type C glenoid), significant osteoarthritis or chondral erosion, neuromuscular abnormality resulting in motor deficit, and patients who have failed a prior arthroscopic procedure (relative contraindication). Patients unable to comply with a postoperative regimen that consists of a brief period of immobilization followed by a structured therapy protocol are not suitable candidates for surgical intervention.

Goals of the Procedure

The main purpose of arthroscopic posterior stabilization is to prevent recurrent macro (dislocation) or micro (RPS) posterior instability of the glenohumeral joint. Traditional techniques focused on open capsular plication with labrum repair, bone block, or wedge osteotomy to prevent recurrent posterior instability. Evolving techniques have allowed for better access to the glenohumeral joint with an arthroscopic approach with accompanying superior clinical outcomes in recent years.¹³ The ultimate goal of arthroscopic management is to restore normal glenohumeral mechanics, prevent recurrent instability or subluxation, and allow for return to normal physiologic function of the joint and athletic participation. The arthroscopic method is also aimed at improving short-term ROM goals and preventing disruption of the deltoid or scapular musculature, which can delay postoperative rehabilitation and may accompany an open technique.

POSTOPERATIVE MANAGEMENT

At the completion of the surgery, the shoulder is placed in a shoulder abduction sling, which immobilizes the shoulder in 30 degrees of abduction and prevents internal rotation. An optional cryotherapy device may be used for the first 3 postoperative days to minimize swelling and pain. During the initial postoperative

Authors' Preferred Technique

Arthroscopic Posterior Shoulder Stabilization

Preoperative Preparation and Positioning

Preoperative Preparation

Prior to surgery, the magnetic resonance arthrogram (MRA) is evaluated for degree of glenoid retroversion, significance of any osseous pathology, and the degree of labrum pathology. Routine radiographs (AP, scapular Y, and axillary view) are also typically obtained prior to MRA to exclude the presence of fracture, subluxation, or frank dislocation. Understanding the needs of each patient is critical to achieving success with this technique. For instance, patients who engage in high-risk contact sports are more likely to have significant labral pathology and require added capsular plication. Patients who are overhead-throwing athletes will have difficulty returning to sports if excessive plication is performed, causing a decrease in ROM and capsular tightness. Patients are given information regarding the risks of surgery, which may include stiffness, infection, neurovascular injury, and recurrence of instability. On the date of the surgery, an interscalene nerve block is routinely used with general anesthesia for both intraoperative muscle relaxation and postoperative pain control.

Positioning

A lateral decubitus position is preferred; however, a beach-chair position can be utilized for this technique. The lateral decubitus position offers excellent access to the posterior joint and capsule without the use of accessory traction by an assistant. A beanbag positioner is typically used with an axillary roll to protect the nonoperative extremity. The down leg is padded over the peroneal nerve, fibular head, and lateral malleolus. The operative arm is placed in 10 to 15 lbs of balanced arm traction in 45 degrees of abduction and 20 degrees of forward flexion. When the posterior, inferior aspect of the labrum is approached arthroscopically, forward flexion can be increased to gain better exposure. The bed should be angled 45 degrees away from anesthesia, and the visualization tower should be placed across the surgical field and at eye level to the surgeon.

Operative Technique

Portal Placement

All bony landmarks are marked with a marking pen (the acromion, coracoid process and acromioclavicular joint). The anterior portal is typically placed in the rotator interval in a trajectory that is diagonal from the coracoid process to the anterolateral edge of the acromion. The posterior portal is created slightly lateral to a traditional posterior arthroscopy portal to allow tangential access to the posterior glenoid for anchor placement. The posterior portal is created first after insufflation of the joint with 30 cc of normal saline. This facilitates safe insertion of the arthroscope cannula and blunt trochar. The anterior portal is created in the rotator interval either using an "inside out" technique with a switching stick, or an "outside in" technique with a spinal needle. Once portals are created, a clear cannula is placed in the anterior portal using a traditional dilation technique. Either a 5.75 mm or 8.25 mm clear cannula (distally threaded) can be used for the anterior portal.

Diagnostic Arthroscopy

A 30-degree arthroscope placed through a posterior viewing portal is used to perform an initial diagnostic arthroscopy (see Figs. 41.2, 41.3). The joint is assessed for concomitant pathology such as superior labral tears, rotator cuff tears, and osteochondral injury. Débridement of the joint can be carried out with a 4.5 mm shaver through the anterior portal as needed. Following an initial diagnostic arthroscopy, the camera is placed into the anterior cannula, and inflow is placed on the side port on the cannula. A switching stick is placed in the posterior portal, and this portal is dilated to 8 mm. An 8.25-mm cannula (Arthrex, Naples, FL) is then placed in the posterior portal. The arthroscope can be switched to a 70-degree arthroscope, which allows better visualization of the posterior and inferior capsule and posterior inferior band of the inferior glenohumeral ligament.

Typical pathology associated with posterior instability includes posterior labral fraying and splitting, posterior labral detachment from the glenoid rim, a patulous posterior capsule, a discrete posterior capsular tear, undersurface partial-thickness rotator cuff tears, and widening of the rotator interval (Fig. 41.10). The surgeon also must be cognizant of the subtle Kim lesion—that is, a concealed incomplete detachment of the posterior labrum.⁵²

Glenoid and Labrum Preparation

A meticulous glenoid preparation is critical to achieving success of the repair. An elevator is initially placed through the posterior portal, and the labrum is sharply elevated off the glenoid (Fig. 41.11). The labrum is mobilized until the underside of the capsule edge of the glenoid rim is visualized. Care is taken during labral elevation and mobilization to not transect the labrum. Adequate trajectory through the posterior portal is critical in achieving this goal. If the angle is difficult, the elevator can be brought in through the anterior portal and the labrum visualized posteriorly. Once adequate mobilization is achieved, a hooded burr is used to prepare the glenoid rim (Fig. 41.12). Great care should be used to ensure the labrum is protected during the bony débridement. A shaver is then used to remove any remaining soft tissue debris from the glenoid rim

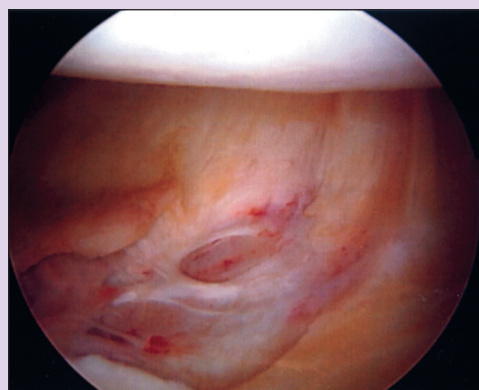


Fig. 41.10 A discrete posterior capsular tear in an athlete with recurrent posterior subluxation of the shoulder (right shoulder, viewed from the anterior portal).



Fig. 41.11 An arthroscopic chisel used to elevate the torn labrum, which is scarred medially, away from the glenoid rim. In this case, a split in the labrum is used to access the scarred tissue (right shoulder, viewed from the anterior portal).

Authors' Preferred Technique

Arthroscopic Posterior Shoulder Stabilization—cont'd

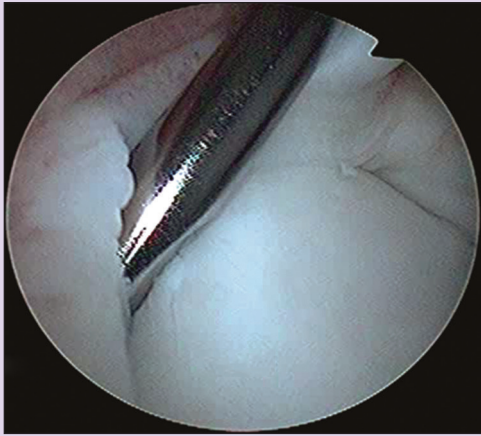


Fig. 41.12 A motorized shaver is used at the posterosuperior extent of the labral tear to decorticate the glenoid rim and abrade the labral undersurface (right shoulder, viewed from the posterior portal). To complete the abrasion posteroinferiorly, the shaver is switched to the posterior portal and the arthroscope is placed in the anterior portal.

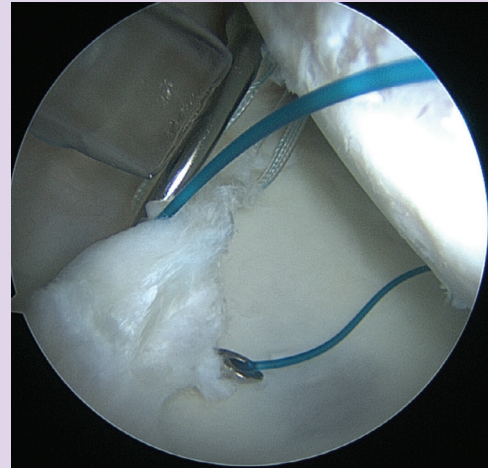


Fig. 41.14 A suture hook device is first used to shuttle a monofilament suture around the labrum for later suture tape passage.

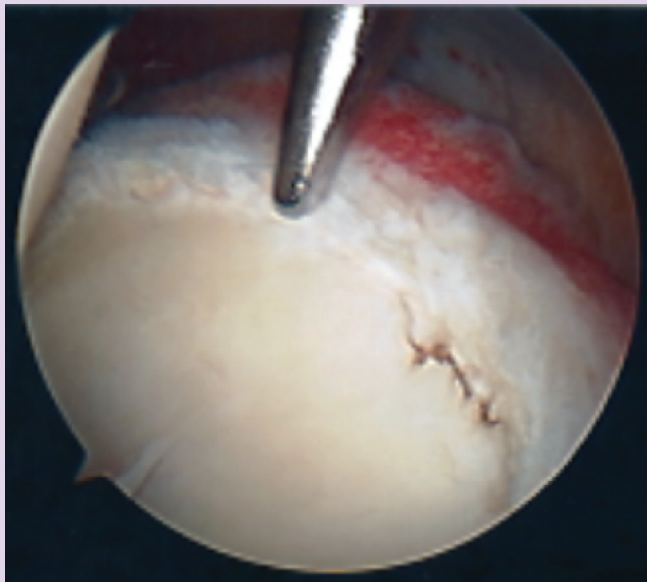


Fig. 41.13 Arthroscopic view from the posterior portal looking inferiorly along the glenoid after labrum mobilization and glenoid preparation is complete.

followed by rasping to create a fresh, bleeding surface conducive to soft tissue healing (Fig. 41.13).

Suture Passing and Labral Repair

A knotless anchor system is employed (2.4 mm biocomposite short PushLock anchors [Arthrex, Naples, FL] for patients with smaller glenoids [subequatorial width <25 mm] or 2.9 mm short PushLock anchors for patients with normal or large glenoid widths) with 1.3 mm Suture Tape (Arthrex, Naples, FL) as our primary fixation method. Knotless fixation superior to the equator of the glenoid may minimize the risk of humeral head chondral injury, as well as articular-sided rotator cuff knot abrasion. Furthermore, the suture tape has a low profile, which

allows it to lay flat against the labrum minimizing tear-through and provides excellent reduction without humeral abrasion. The first step in labral repair is predicated on the surgeon understanding the specific pathology and demands of the patient. Patients presenting with symptoms consistent with RPS and insidious shoulder pain without gross macro-instability (i.e., throwers) may benefit from a primary labral repair without significant capsular plication/advancement. Patients with macro-instability or dislocation and who participate in high-demand contact sports (i.e., offensive lineman) may require labral repair with plication of the posterior capsule.

For the standard posterior labral repair, we prefer to place anchors from approximately 6-o'clock to the 11-o'clock position on the glenoid. To achieve positioning of the inferior most anchor, an accessory posterior portal may need to be created 45 degrees tangential to the glenoid. This portal is typically created slightly inferior and lateral to the posterior working portal. While viewing from the anterior portal, a spinal needle is used to localize the correct site of entry for the accessory posterior portal. Once the correct location and trajectory are identified for anchor placement, an accessory 5 mm cannula is introduced into the joint using Seldinger technique. Great care should be taken to avoid damage to the axillary nerve on cannula introduction, as it lies at risk in this area. The anchors are all placed from inferior to superior after suture passage.

A curved suture hook (Spectrum, Linvatec, Edison, NJ, or ReelPass Suture Lasso, Arthrex, Naples, FL) loaded with a 0 monofilament suture is placed through the posterior working cannula while the arthroscope is placed in the anterior cannula. A bite of capsulolabral tissue is taken slightly inferior and lateral to the level of desired anchor location (to achieve capsular plication and advancement) and the suture is delivered through the labrum against the glenoid articular margin (Fig. 41.14). The monofilament suture is then shuttled through the posterior working cannula, and a loop of Suture Tape is placed in a knot of the monofilament suture. The goal is to create a "cinch" stitch akin to a luggage tag that will reduce the labrum and provide optimum suture security around the tissue. The loop of Suture Tape is then shuttled back through the posterior working cannula, and the tail ends of the suture are delivered through the loop created in the Suture Tape. The two tails of the Suture Tape are then pulled, and the cinch stitch is reduced to the labrum. A knot pusher can be used to help reduce the knot to the labral tissue if needed when there is concern that the tissue is of poor quality. This creates 2.6 mm of Suture Tape, as the tape is essentially doubled over to create a stable fixation construct around the labrum and capsular

Continued



Authors' Preferred Technique

Arthroscopic Posterior Shoulder Stabilization—cont'd

complex. The Suture Tape ends are then delivered through the accessory cannula for anchor placement. The drill guide is then placed through the accessory posterior cannula and a pilot hole drilled in the desired location. The suture ends are then shuttled through the eyelet of the knotless anchor and the anchor is then slid to the level of the glenoid and slight tension is applied to the limbs to reduce the labrum (Fig. 41.15). The anchor is then impacted into place within the pilot hole and the labrum, and capsule is reduced to the glenoid margin (Fig. 41.16). Additional anchors can be placed in an identical fashion as needed. Anchors in the 8-o'clock to 11-o'clock position are generally placed through the standard posterior portal, which provides a better angle of approach in this zone.

Posterior Portal Closure

The posterior portal is then closed using a No. 1 Polydioxanone (PDS) suture. The cannula is withdrawn to a level just external to the capsule. A suture passer

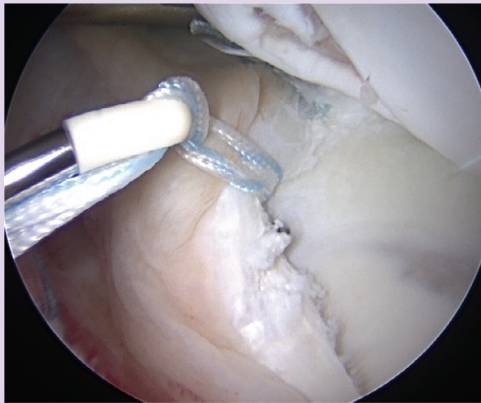


Fig. 41.15 A luggage tag or "cinch" stitch has been placed around the labrum and capsule and is delivered through the eyelet of the anchor for glenoid placement. Note the osteochondral injury present in this patient on the posterior aspect of the humeral head.

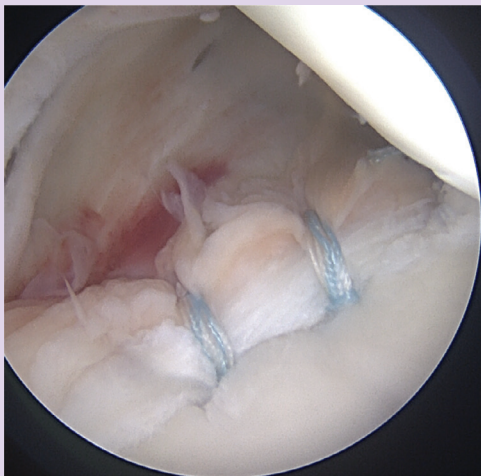


Fig. 41.16 Arthroscopic view from the posterior portal looking inferiorly at the completed labrum and capsular repair.

is then used to pierce the capsule adjacent to the portal, and the PDS suture is introduced into the joint. A penetrating suture grasper is then used to pierce the capsule on the opposite side of the portal, and the suture is retrieved. A sliding Weston knot is then used to close down the capsule. Additional sutures can be placed in a wider fashion if additional plication is required based on the patient's predetermined needs from the preoperative plan.

Technical Considerations

- **Positioning**—The lateral decubitus position offers excellent exposure with minimal assistance to access the posterior joint capsule.
- **Portal Placement**—Placement of the posterior portal slightly lateral to the standard portal allows for easier placement of the anchors when utilizing a single portal technique.
- **Anchor Placement**—Anchors should be placed nearly perpendicular to the glenoid or at a tangent of 45 degrees to avoid chondral injury and "skiving."
- **Repair**—The repair should be tailored to the precise pathology of the patient as determined by history, examination, and diagnostic imaging. Knotless techniques can help avoid knot abrasion against the rotator cuff and scuffing against articular cartilage (Fig. 41.17).

What to Avoid

- Avoid placement of anchors too close to one another, as this could cause fragmentation of the glenoid bone and a "postage stamp" style fracture.
- Avoid transecting the posterior labrum during labrum preparation. Use of a periosteal elevator should be done from the portal that allows for the most mobilization of tissue without iatrogenic injury.
- Avoid over-tightening overhead-throwing athletes, as this could compromise their ability to RTP.

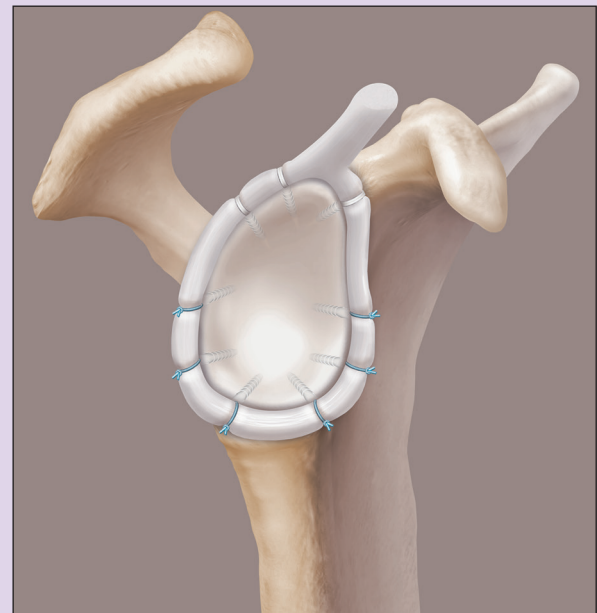


Fig. 41.17 Knotless anchors used for labral repair above the horizontal equator of the glenoid (right shoulder), with traditional suture anchors used below. (Courtesy of Arthrex Inc., Naples, FL.)

period, patients begin active ROM of the elbow, wrist, and fingers. At 1 week after surgery, formal physical therapy for the shoulder is initiated, beginning with passive forward flexion and abduction in the scapular plane to 90 degrees. During the subsequent 5 weeks, full passive ROM is achieved. At 6 weeks after surgery, use of the sling is discontinued and active-assisted ROM begins, progressing to full active ROM according to the patient's ability. Most patients are progressed to full active ROM by 8 to 12 weeks. Strengthening of the rotator cuff, deltoid, and periscapular muscles begins at 3 months after surgery.

Once patients are able to achieve 80% of the strength of their contralateral shoulders as measured by isokinetic testing, a sport-specific rehabilitation protocol is initiated that typically takes place between 4 and 6 months after surgery. Athletes are cleared to return to sport when they demonstrate full ROM and strength, along with restoration of glenohumeral stability, typically between 6 and 9 months after surgery.

Throwing athletes require special consideration and follow a specific protocol designed to monitor their throwing distance and speed, which is slowly advanced over 2 to 3 months. Initially, at 6 months after surgery, an easy tossing program is begun at a distance of 20 ft, without a windup. Before each session, stretching is performed and heat is applied to increase circulation and improve ROM. At 7 months, light throwing with an easy windup at a distance of 30 ft is allowed 2 to 3 days per week for 10 minutes per session. Easy throws from 150 to 200 ft are permitted 9 months after surgery, and stronger throws from the same distance are allowed at 10 months. Pitchers can throw at half to three-quarters speed from the mound, with an emphasis on accuracy and technique, at 11 months. At 12 months, pitchers can throw at three-quarters to full speed. Throwers are allowed to return to full competition when they are able to throw at full speed without discomfort for 2 weeks, typically between 9 and 12 months after surgery.

RESULTS

With expanding interest in minimally invasive surgical techniques, the first successful accounts of arthroscopic posterior glenohumeral stabilization were produced in the 1990s.^{71–73} In 1998, Wolf et al. reported on 14 patients who had arthroscopic posterior capsular plication for unidirectional posterior instability; at 33-month follow-up, 86% had good or excellent results and 93% had restoration of stability.⁷³ With improvement in arthroscopic techniques and surgical implants, research has led to the evolution of suture-anchor–based arthroscopic posterior glenohumeral stabilization. Numerous recent studies have revealed that good or excellent results have been achieved in a majority of cases, driving further interest in these methods.

Arthroscopic stabilization for unidirectional posterior glenohumeral instability has been of particular interest. Williams et al. retrospectively reported on 27 shoulders with symptomatic posterior capsulolabral complex detachment from the posterior glenoid rim, with minimal posterior capsular laxity, after a distinct traumatic event.¹⁴ Isolated arthroscopic posterior capsulolabral repair was performed with bioabsorbable tack fixation in all patients.¹⁴ Subjective instability and pain were eliminated in 92%

of patients, with all patients returning to unlimited athletic activity by 6 months. Kim prospectively reported on 27 athletes with traumatic unidirectional RPS who were treated with arthroscopic posterior labral repair and capsular shift; all patients had a labral lesion, and 81% also had stretching of the posterior band of the IGHL.⁵² The posterior capsule was shifted superiorly in all cases. All complete labral lesions were repaired directly, and incomplete labral lesions were converted to complete tears and then repaired. The mean ASES scores improved from 51.2 to 96.5. All patients except for one experienced restoration of stability and were able to return to their previous athletic activities with few or no limitations.

Multiple additional studies involving between 12 and 34 shoulders have revealed similar successful results for arthroscopic posterior stabilization, with recurrence rates between 0% and 12% and a minimum mean follow-up of 3 years.^{6,8,36,74–77} In 2008, Radkowski et al. reported on arthroscopic capsulolabral repair for posterior shoulder instability in throwing athletes compared with nonthrowing athletes.⁷⁴ At 27-month mean follow-up, no differences were found in the ASES score or scores for stability, ROM, strength, pain, and function between the 27 throwing shoulders and the 80 nonthrowing shoulders, with both groups showing significant improvement in all categories. Excellent or good results were achieved in 89% of throwers and 93% of nonthrowers. However, throwing athletes were less likely to return to the preinjury level of sport (55%) compared with nonthrowing athletes (71%).⁷⁴ Due to the relatively small patient population in this study, the reason for the lower rate of return to sport in throwers was difficult to identify. Interestingly, all of the three throwers in whom the procedure failed underwent capsulolabral repair without suture anchors. Based on the findings of this study, the authors presently advocate repair with suture anchors uniformly. In addition, within the thrower group more athletes were unable to return to their previous levels of sport compared with those for whom the procedure failed based on the subjective instability scale, which suggests that some of these athletes might have had lower levels of subjective instability but enough to keep them from returning to the same high level of competition and meeting the unique demands required from the shoulder during throwing activities.

In 2008, Savoie et al. reported a 97% success rate for the treatment of 92 shoulders with posterior instability using arthroscopic techniques.⁷⁸ The mean age of the patients was 26 years, and mean follow-up was 28 months. Pathology discovered at the time of surgery was an isolated reverse Bankart lesion in 51%, a stretched posterior capsule in 67%, and a combination of a reverse Bankart lesion and capsular stretching in 16%. The rotator interval was determined to be damaged in 61% and was closed in all but five patients (95%). One patient had recurrent gross subluxation, and one had recurrent dislocation.

Lenart et al. reported 100% return to previous level of athletic activity for 34 shoulders for which arthroscopic repair for posterior shoulder instability was performed.⁷⁷ The majority (78%) had sustained a traumatic event, but only two (6%) entailed a frank dislocation, consistent with a population of RPS. Suture anchor repair was performed in 30 patients (88%), and capsular plication to an intact labrum was performed in four patients (12%).

Wanich and colleagues also presented the results of arthroscopic treatment of posterior capsulolabral lesions in persons with batterer's shoulder, defined as posterior subluxation of the lead shoulder during a baseball swing.³⁶ Eleven of 12 patients treated with surgery (92%) returned to their previous level of batting at an average of 5.9 months after surgery, with no detectable ROM deficit compared with before surgery.

Arner et al. evaluated 56 consecutive American football players with posterior shoulder instability who underwent an arthroscopic posterior capsulolabral repair with or without suture anchors.⁷⁹ At a mean follow-up of 44.7 months, 93% returned to sport and 79% returned to the same level of play. Statistically significant improvements were seen in ASES scores, as well as scores of stability, ROM, strength, pain, and function. Excellent or good results (ASES score >60; stability <6) were achieved in 96.5% of athletes, and 96% were satisfied with their surgery.⁷⁹

In 2006, Bradley presented a larger prospective study that included 91 athletes (100 shoulders) treated with arthroscopic capsulolabral reconstruction for posterior instability.⁸ He expanded upon this initial study to include 183 athletes (200 shoulders) in 2013, which is the largest patient cohort to date.⁸⁰ Included in the study were 99 athletes involved in contact sports and 101 athletes involved in noncontact sports. All patients had unidirectional posterior instability, and a course of nonoperative management had failed. Arthroscopy with suture anchor repair was the most common procedure performed ($n = 119$), followed by suture-only repair without anchors ($n = 44$) and suture anchor repair with supplementary plication sutures ($n = 37$). In his follow-up study at a mean of 36 months postoperatively, the mean ASES scores improved from 45.9 to 85.1, with 90% return to sport and 64% returning to their same level as before surgery. Potential reasons elucidated for the failures included undiagnosed MDI, inadequate capsular shift because of unappreciated capsular laxity, inadequate recognition of poor capsular tissue quality in patients referred after thermal capsulorrhaphy, and capsular plication performed in an isolated fashion without suture anchors.^{8,80} The authors also concluded that the incorporation of bone suture-anchors in capsulolabral reconstruction led to higher ASES scores and rate of RTP.⁸⁰

COMPLICATIONS

The most likely complication resulting from arthroscopic posterior stabilization is recurrent instability (<10%). The majority of patients, however, will achieve a successful outcome with proper technique and adherence to a strict protocol-driven rehabilitation program. Postoperative stiffness is relatively uncommon but can be encountered in patients that underwent concomitant posterior capsular plication. Early posterior capsular stretching can help avoid this complication, and in those patients where ROM is not restored, posterior capsulotomy can be performed if deemed necessary.

Various intraoperative complications may occur during arthroscopic posterior stabilization. Erroneous placement of the 7-o'clock posterior portal may result in damage to the posterior humeral circumflex artery or the suprascapular nerve and artery,

which lie on average 39 mm and 29 mm away, respectively.⁸¹ The axillary nerve is also at risk during arthroscopic repair of the posteroinferior shoulder capsule, particularly with placement of the most inferior suture anchor. In a cadaveric study, Price et al. performed a microsurgical dissection through the IGHL from within the joint capsule to examine the relationship of the axillary nerve as it traversed the quadrangular space.⁸² In each dissection, the teres minor branch was the closest to the glenoid rim. The closest point between the axillary nerve and the glenoid rim was at the 6-o'clock position on the inferior glenoid rim. At this position, the average distance between the axillary nerve and the glenoid rim was 12.4 mm. The axillary nerve lay, throughout its course, at an average of 2.5 mm from the IGHL. Thus when performing arthroscopic plication on the inferior glenohumeral joint capsule, care must be taken to avoid overpenetration with the suture passer, which could injure the adjacent axillary nerve. Further, the use of a thermal device for inferior capsular shrinkage should be avoided, because axillary nerve injury has been reported with that surgical technique.⁸³

Perioperative complications can also occur as result of the implants selected for arthroscopic stabilization. Metallic anchors were first produced and used in soft tissue fixation around the shoulder. However, their use resulted in some reported complications, including articular surface damage from migrating implants and distortion, and artifact production in postoperative MRI.⁸⁴ Bioabsorbable anchors were developed to avoid these problems and have proved to have pull-out strength equal to that of metallic anchors, with a reported lower complication rate.⁸⁴ This finding led to a major shift away from metallic anchors toward bioabsorbable anchors. However, the use of some bioabsorbable suture anchors in the shoulder—in particular, those composed of polylactic acid—have resulted in notable complications. Rare cases of premature absorption, osteolysis, glenohumeral synovitis, cyst formation, soft tissue inflammation, and release of implant fragments into the joint space have been reported.^{85–89} These results have been thought to be due specifically to the biologic properties of these early-generation implants. Thus patients who experience postoperative pain, stiffness, or mechanical symptoms or fail to progress as expected should be examined for these potential implant-related complications. Recent improvements in bioabsorbable suture-anchor materials, namely concerning their mechanical and resorptive properties, have potentially led to a decrease in these biocompatibility issues; further study is warranted to confirm these observed trends. In addition, newer materials, such as biocomposites and polyetheretherketone (PEEK), have recently been introduced. These materials may address concerns of biocompatibility and material strength, but additional rigorous in vitro and in vivo trials need to be conducted before their use becomes widespread.⁸⁸

Iatrogenic glenoid chondral damage is a risk during arthroscopic posterior labral repair and typically occurs with a low, tangential drilling angle (<30 degrees) during anchor placement. This complication can be avoided with proper placement of the posterior portal, which allows a higher angle (>30 degrees) of approach to the glenoid rim. Transection of the labrum during preparation can occur and compromise the

ability to achieve an anatomic repair. In instances where this occurs, the posterior capsule can be used utilizing an extracapsular technique to repair the capsule to the posterior glenoid margin. In addition to chondral or labral damage, malposition of the anchors can also lead to pain, decreased ROM, and failure of the repair.⁹⁰

Venous thromboembolism (VTE) events are rare, but not unheard of, after shoulder arthroscopy. In a recent analysis of more than 65,000 arthroscopic procedures, Jameson et al. found the rate of deep venous thrombosis, pulmonary embolism, and mortality rates to be less than 0.01%, 0.01%, and 0.03%, respectively.⁹¹ The authors concluded that thromboprophylaxis for VTE events may not be required, even in high-risk patients, because the 2007 National Institute for Health and Clinical Excellence guidelines did not affect VTE event rates.

SUMMARY

Posterior shoulder instability poses a significant challenge to the orthopedist, both diagnostically and therapeutically. It is a broad clinical entity, ranging from RPS to locked posterior dislocations. Pathogenesis also varies tremendously and is based on the stress placed on the involved shoulder. As opposed to a singular traumatic event, athletes with RPS typically experience chronic, repetitive microtrauma that leads to a posterior labral tear, posterior capsular laxity, and symptoms. Overhead throwers are a unique patient population, often with a varied presentation that can include posterior capsular contracture and the type VIII SLAP tear. Extensive basic science and clinical research is ongoing to elucidate a biomechanical explanation of the pathogenesis of RPS in throwers. Increased clinical awareness, as well as advances in imaging and physical examination techniques, have improved diagnosis in athletes and patients from wide demographic presentations. Multiple distinct pathologic lesions can be identified on MRA in patients with RPS; these lesions should be sought carefully during the workup of symptomatic patients. Specific provocative physical examination tests have been found to be useful in making the diagnosis of posterior shoulder instability.

Successful outcomes can be achieved with physical therapy, although many patients still require surgical intervention to alleviate symptoms and return to sport at their preinjury level. The transition from open to arthroscopic surgical techniques has facilitated more comprehensive identification and treatment of coexisting structural pathologies in persons with RPS. For patients with unidirectional posterior instability, an arthroscopic repair of the capsulolabral complex is recommended. When a patulous posterior capsule or a posteroinferior component of instability is also present, an arthroscopic superiorly directed capsular shift and capsular plication are added. In throwing athletes with RPS and a posterior labral tear, we advocate repairing the posterior labrum only and initially treating excessive GIRD, if present, with preoperative sleeper stretches and, if it is recalcitrant, with a limited posterior capsular release at the time of surgery. With regard to addressing capsular pathology, we generally tend toward overtightening in athletes in contact sports and in persons with significant capsular laxity and undertightening in overhead throwers.

FUTURE CONSIDERATIONS

Success rates between 90% and 100% have been achieved with arthroscopic repair techniques, with generally lower results in throwing athletes. With midterm results demonstrating favorable results, longer term studies are necessary to evaluate the durability of arthroscopic techniques in the treatment of posterior glenohumeral instability. Further work is also needed to elucidate the complex mechanics of the elite thrower's shoulder and the best path to alleviating the symptoms of RPS while simultaneously allowing return to sport at the preinjury level.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:

DeLong JM, Jiang K, Bradley JP. Posterior instability of the shoulder: a systematic review and meta-analysis of clinical outcomes. *Am J Sports Med*. 2015;43(7):1805–1817, 2015.

Level of Evidence:

IV

Summary:

This systematic review and meta-analysis reviews the literature regarding clinical outcomes for arthroscopic and open treatment of unidirectional posterior instability of the shoulder. Fifty-three unique publications were included. Despite overall similar results in the overall athletic population, throwing athletes are less likely to return to preinjury level of play. Literature suggests arthroscopic treatment has improvement in outcomes in regard to recurrence of instability, patient satisfaction, return to sport and preinjury level of play.

Citation:

Kim SH, Ha KI, Park JH, et al. Arthroscopic posterior labral repair and capsular shift for traumatic unidirectional recurrent posterior subluxation of the shoulder. *J Bone Joint Surg Am*. 2003;85A(8):1479–1487.

Level of Evidence:

II

Summary:

This clinical study features the Kim classification for describing posterior labral tear morphology, including the “Kim lesion,” and a report on the overall successful arthroscopic management of unilateral recurrent posterior subluxation in 37 patients.

Citation:

Song DJ, Cook JB, Krul KP, et al. High frequency of posterior and combined shoulder instability in young active patients. *J Shoulder Elbow Surg*. 2015;24(2):186–190.

Level of Evidence:

III

Summary:

This retrospective review of 231 consecutive patients evaluated the frequency of posterior and combined shoulder instability in patients that underwent operative stabilization procedures. Secondly this study assessed preoperative magnetic resonance imaging reports to determine their accuracy in identifying pathology that was addressed at the time of surgery.

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